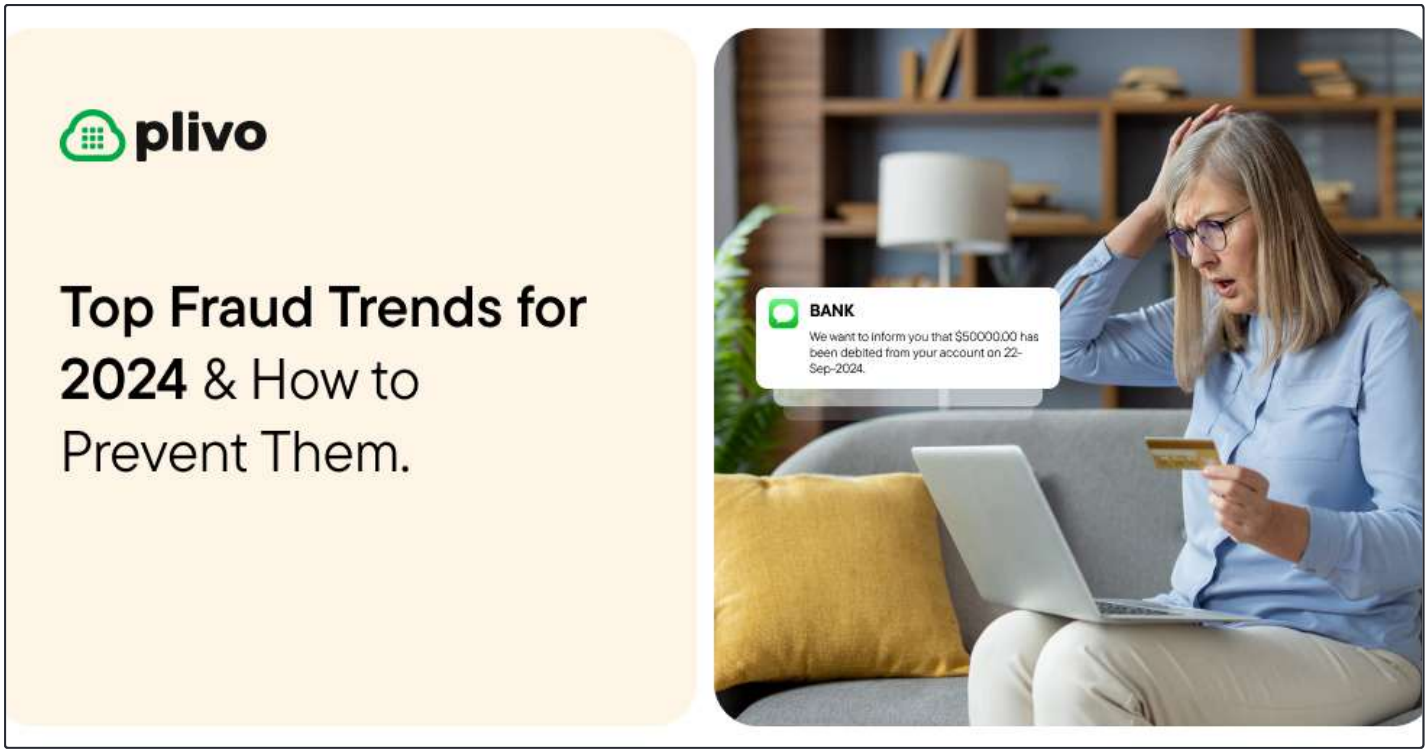
SIGN UP FOR FREE

Fraud Prevention

Understanding One-Time Passcode Fraud

Learn what OTP fraud is and how to protect yourself from online scams using one-time passwords. Stay safe with simple tips and prevention methods.

November 26, 2024 · By Team Plivo



One-time password (OTP) scams are a growing threat to businesses and consumers alike.

In 2021, US consumers lost a staggering $13.9 billion to fraud, with OTP scams accounting for a significant portion of these losses. This rising trend poses serious financial and reputational risks for businesses.

But what exactly is OTP fraud, and how do scammers exploit it? Let's see.

SIGN UP FOR FREE

An OTP is sent to individuals by mobile phone or email. It's used to authenticate transactions such as logging into accounts or making payments.

Fraudsters, however, exploit this security feature through various tactics. These include SIM swapping, phishing emails, fake tech support calls, and social engineering. They design each of these methods to trick individuals into revealing their OTPs.

How does OTP fraud work?

OTP fraud takes place when an attacker uses the OTP system to gain unauthorized access to someone's financial accounts or online profiles. Here's how it works:

1. **Targeting victims:** Fraudsters often select targets based on their online behavior, weak points in communication methods like SMS or email, or publicly shared information on social media.
2. **Launching the attack:** They employ phishing techniques to impersonate legitimate organizations, such as banks or e-commerce platforms. They focus on creating a sense of urgency through emails, SMS, or phone calls to trick targets into taking action.
3. **Collecting personal information:** The phishing messages lead to fake websites that look like trusted ones. Here, targets unknowingly provide personal details like usernames and passwords.
4. **Trapping with OTPs:** The fake website triggers the OTP and sends it to the target's phone. Fraudsters then use various methods to steal the OTP, such as:

    - **Social engineering:** The attacker may try to convince the individual to share the OTP, claiming it is for account verification.
    - **Malware Interception:** If malware infects the individual's device, it intercepts and captures the OTP without the user's knowledge.

SIGN UP FOR FREE

steal data, or take control of future OTPs.

Common methods used by scammers

OTP phishing attacks take many forms, each targeting different vulnerabilities in users.

Below are some common OTP fraud tactics:

Smishing

Smishing is a term that combines “SMS” and “phishing.” Here, fraudsters use fake text messages to trick individuals into downloading malware or sharing sensitive information, lik OTPs.

Smishing attacks have become more common. In 2023, 75% of organizations reported faci these types of attacks.

Scammers often hide their identity by spoofing phone numbers or sending texts from email accounts. For example, smishing texts often include links or attachments that seem safe but lead to fake websites.

These imitation sites look like real financial or service provider portals, where users may unknowingly enter their login credentials and OTPs. This gives scammers access to real accounts.

SIM swapping

SIM swapping occurs when fraudsters trick a mobile carrier into transferring a victim's pho number.

Here’s how it works:

- Attackers gather personal information like names, addresses, or answers to security questions, often through phishing, data breaches, or dark web

SIGN UP FOR FREE

carrier, claiming that the scammers lost or damaged the SIM card. They then ask the carrier to transfer the victim's phone number to a new SIM.

- If the carrier fails to verify the request, the fraudster takes control of the number. From there, they intercept calls, texts, and verification codes, allowin them to bypass security and access sensitive accounts.

This growing problem has had a significant financial impact.

In 2023 alone, SIM swapping fraud caused over $48 million in losses, targeting both individuals and companies

So, to protect your business from SIM-swapping attacks, combining the right technology wi a proven approach is essential.

Using Plivo, you can validate phone numbers smoothly, ensuring fraudsters can't take advantage of SIM swaps.

The Plivo Lookup API helps detect SIM swaps in real time. It offers phone number validation and key analytics to assess risks. Some of the critical data it provides includes:

- Current and original network details
- Roaming status and network changes
- Risk scores and unusual patterns that indicate fraud

Businesses can spot suspicious activity and take action before any harm occurs with these insights.

SMS pumping

SMS pumping, or artificially inflated traffic (AIT), is a cybercrime aimed at businesses that re on SMS for sending OTPs or app download links.

SIGN UP FOR FREE

messages to these fake numbers.

For businesses, SMS pumping increases messaging costs, wastes resources, and can overwhelm systems.

Plus, if customers see too many unnecessary OTPs, it may damage their trust in the company.

To address this, Plivo has two built-in security layers—Fraud Shield and SMS Pumping Protection—that help protect Verify and Messaging traffic from SMS pumping attacks. Both features use an advanced detection model to identify artificially inflated traffic and block suspicious messages before they're sent, helping reduce exposure to fraud and unnecessa costs.

The Geo Permissions feature also gives you control over where messages are sent by allowing you to enable or disable countries and apply message-per-hour thresholds based your risk tolerance.

Voice phishing (vishing)

Vishing, or voice phishing, is a type of scam where fraudsters use phone calls to trick peop into sharing sensitive information, like login details, credit card numbers, or bank account info. Scammers exploit this data for identity theft and financial fraud, making vishing widespread and costly.

In 2023, over 56 million U.S. adults — about 21% of the population — fell victim to phone scams, resulting in losses totaling $25.4 billion.

In vishing scams, attackers often impersonate trusted organizations and use phone calls to create a sense of urgency and trust. They also build a personal connection over the phone, reacting to what the victim says and using emotional cues.

How OTP scams impact businesses

Here's how scammers misuse OTPs and why this should be a major concern for businesses

SIGN UP FOR FREE

Stealing or tricking users into sharing their OTP allows fraudsters to bypass security, gain control of accounts, and access private information or funds.

This can harm trust and cause legal problems for businesses, especially those handling sensitive data or finances.

In fact, the average cost of a data breach in 2023 was $4.45 million, underscoring the serious financial impact of such security failures.

Unauthorized transactions

With OTPs, scammers can authorize fraudulent transactions. If they gain access to a customer's account, they can make purchases, transfer funds, or exploit stored payment methods.

As a result, businesses may face chargebacks, financial losses, and increased scrutiny from payment providers due to such fraudulent activities.

Fake account registration and spamming

Some scammers use stolen OTPs to create fake accounts on business platforms, taking advantage of promotional offers or testing one-time passcode security.

Fake registrations can distort the numbers, affect data accuracy, and lead to spam, resulting in unnecessary management overload.

Spreading malware or phishing attacks

If scammers access your business email or social media accounts, they can send phishing messages or malware links to your contacts.

Scammers make these messages look authentic, so recipients are more likely to trust and open them. This spreads the scam further and damages your reputation. It also puts your clients, partners, and employees at risk.

SIGN UP FOR FREE

To effectively protect your business from OTP fraud, it's important to take a proactive approach. Implementing strong security measures can significantly reduce risks.

Here are some practical tips for businesses to safeguard against OTP fraud:

Implement 2FA with fraud prevention

Two-factor authentication (2FA) adds an important layer of security by requiring both a password and an OTP.

However, despite its effectiveness, many organizations are still not fully adopting this secur measure. In fact, a recent survey found that 64% of organizations haven't fully adopted multi-factor authentication (MFA).

This is concerning because research shows that MFA can block over 99.9% of account takeover attacks. Without MFA, businesses are leaving their accounts — and their customer — vulnerable to fraud.

Plivo's 2FA/OTP capabilities can help overcome OTP verification vulnerabilities. Adding thes extra layers of security prevent account takeovers, reduce fraud risks, and secure high-valu transactions.

Additionally, Plivo provides several advanced features to further enhance security:

- **Geo permission management:** Control which countries your Plivo account ca send SMS or make calls to, blocking high-risk areas.
- **International toll fraud protection:** Manage connections to premium numbers and set blacklists to prevent toll fraud.
- **Pattern-based alerts:** Get notified about unusual SMS or voice activity, like sudden drops in delivery rates, to detect potential fraud early.

Given the growing risk of cyberattacks, it’s more important than ever to implement 2FA. By doing so, you can better protect your business and customers from fraud.

SIGN UP FOR FREE

Delays or failed OTP deliveries can frustrate users, damage trust, and open doors for attackers to exploit system weaknesses.

With Plivo's real time delivery reports, businesses can monitor OTP performance globally. This helps track delivery rates and identify potential issues.

Businesses can use this data to quickly address problems and ensure they deliver OTPs promptly and securely.

Limiting OTP validity and attempt frequency

Setting a time limit for OTP validity and limiting the number of attempts helps prevent misuse and reduces fraud risk.

Businesses can also lower the chances of attackers guessing or intercepting valid codes. This ensures that only legitimate users can complete the authentication process within the allowed time.

Regularly update your security protocols and stay compliant

Staying compliant with security standards ensures that businesses are using the latest practices to protect data. This helps guard against new fraud techniques and reduces the risk of breaches.

Choose a OTP service provider, such as Plivo, that ensures continuous compliance with security standards. Plivo designs its OTP solutions to adhere to regulations like the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA).

With advanced security features and detailed reporting, Plivo makes it easier for businesses to stay compliant and protect sensitive user data.

Choose a secure algorithm

Use strong algorithms like HMAC-based one-time passwords (HOTP) or time-based one-time passwords (TOTP).

SIGN UP FOR FREE

Provide multiple delivery options

Flexibility is important for OTP delivery. Offering choices like SMS, email, or voice ensures users can receive their OTPs in the way that suits them best.

This also improves user experience and offers a backup if one method doesn't work.

Educate users on security best practices

Teaching users about security best practices for OTPs reduces the risk of fraud and strengthens overall protection. Encourage users to create strong, unique passwords, avoid sharing OTPs, and recognize phishing attempts.

Regularly remind them about the importance of safeguarding their accounts and following secure authentication processes to prevent unauthorized access.

Prevent OTP fraud with the help of Plivo today

When selecting an OTP provider, it's important to choose one that offers both robust securit and user-friendly features. Plivo provides a secure and adaptable OTP solution, designed to fit your unique needs.

Plivo's OTP services simplify the authentication process, offering global SMS delivery, instan reporting, and customizable templates for easy implementation.

Whether it's safeguarding transactions, verifying users, or securing sensitive information, Plivo's OTP services offer the reliability and flexibility necessary for your security needs.

Want to see how Plivo can assist in preventing OTP fraud? Contact us today.

Team Plivo

SIGN UP FOR FREE

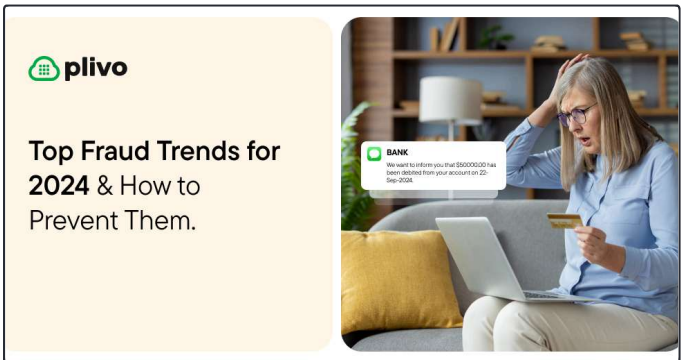
Related Articles



Fraud Prevention

What is SIM Swapping Fraud and How to Prevent It

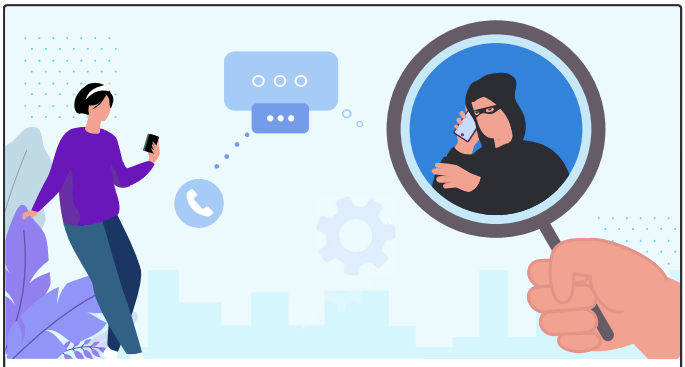
Nov 11, 2024



Fraud Prevention

Top Telecom Fraud Trends for 2024 & How to Prevent Them

Oct 8, 2024



Fraud Prevention

Plivo's Guide to Toll Fraud Prevention: How to Protect Your Business

Jul 5, 2024

SIGN UP FOR FREE

PLATFORM

AI Agents

SIP Trunking

Audio Streaming

Phone Numbers

Pricing

COMMUNICATIONS

Voice

SMS

WhatsApp

Verify

DEVELOPERS

SIP Trunking guide

Audio Streaming quickstart

Documentation

RESOURCES

Platform status

LiveKit

Pipecat

Compliance

COMPANY

About

Jobs

Talk to sales

Get support

Legal

© 2026 Plivo Inc.